Chapter 5, Section 6

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1. Let X be a surface in $\mathbb{P}^n$, $n \geq 3$, defined as the complete intersection of hypersurfaces of degree $d_1, \ldots, d_{n-2}$, with each $d_i \geq 2$. Show that for all but finitely many choices of $(n, d_1, \ldots, d_{n-2})$, the surface X is of general type. List the exceptional cases, and where they fit into the classification picture.

Proof. We will use (II, Ex. 8.4e), which states the canonical sheaf ω_X is isomorphic to $\mathcal{O}_X(\sum_{i=1}^{n-2} d_i - n - 1)$, and assume d_i is in descending order. Since $d_i \geq 2$, for every $(n, d_1, \ldots, d_{n-2})$ with $n \geq 6$, the inequality

$$\sum_{i=1}^{n-2} d_i - n - 1 > 0$$

always holds, which implies ω_X is very ample, hence X is of general type. If $n = 3$ and $d_1 \geq 5$, then X is of general type. If $n = 4$ and either $d_1 \geq 4$ or $d_2 \geq 3$, then X is of general type. If $n = 5$ and $d_1 \geq 3$, then X is of general type. Thus, the exceptional cases are:

$\boxed{n = 3, d_1 = 2}$ X is the nonsingular quadric surface, isomorphic to $\mathbb{P}^1 \times \mathbb{P}^1$, and it is also the rational ruled surface with invariant $e = 0$.

$\boxed{n = 3, d_1 = 3}$ X is a cubic surface, which is isomorphic to $\mathbb{P}^2$ with 6 points in general position blown up, and it is also called a Del Pezzo surface of degree 3.

$\boxed{n = 3, d_1 = 4}$ In this case, $\omega_X \cong \mathcal{O}_X$, and we have a short exact sequence

$$0 \longrightarrow \mathcal{O}_{\mathbb{P}^3}(-4) \longrightarrow \mathcal{O}_{\mathbb{P}^3} \longrightarrow \mathcal{O}_X \longrightarrow 0 \ ,$$

and the long exact sequence in cohomology gives

$$\mathrm{H}^1(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}) \longrightarrow \mathrm{H}^1(X, \mathcal{O}_X) \longrightarrow \mathrm{H}^2(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(-4)) \ ,$$

so $q = h^1(X, \mathcal{O}_X) = 0$. Hence, X is a K3 surface.

$\boxed{n = 4, d_1 = d_2 = 2}$ X is a Del Pezzo surface of degree 4 with $\omega_X \cong \mathcal{O}_X(-1)$, which is isomorphic to $\mathbb{P}^2$ with 5 points blown up (Ex. 4.13).

$\boxed{n = 4, d_1 = 3, d_2 = 2}$ In this case, $\omega_X \cong \mathcal{O}_X$. Let $\mathcal{I}$ be the ideal sheaf of X in $\mathbb{P}^4$. The long exact sequence in cohomology gives the isomorphism

$$\mathrm{H}^1(X, \mathcal{O}_X) \cong \mathrm{H}^2(\mathbb{P}^4, \mathcal{I}).$$

By (III, §7), we can use the Koszul resolution of $\mathcal{I}$ to compute its cohomology

$$0 \longrightarrow \mathcal{O}_{\mathbb{P}^4}(-5) \longrightarrow \mathcal{O}_{\mathbb{P}^5}(-3) \oplus \mathcal{O}_{\mathbb{P}^4}(-2) \longrightarrow \mathcal{I} \longrightarrow 0 \ ,$$

which readily shows $\mathrm{H}^2(\mathbb{P}^4, \mathcal{I}) = 0$. Hence, X is a K3 surface.

$\boxed{n = 5, d_1 = d_2 = d_3 = 2}$ In this case, $\omega_X \cong \mathcal{O}_X$. Let $\mathcal{I}$ be the ideal sheaf of X in $\mathbb{P}^5$. The long exact sequence in cohomology gives the isomorphism

$$\mathrm{H}^1(X, \mathcal{O}_X) \cong \mathrm{H}^2(\mathbb{P}^5, \mathcal{I}).$$

By (III, §7), we can use the Koszul resolution of $\mathcal{I}$ to compute its cohomology

$$0 \longrightarrow \mathcal{O}_{\mathbb{P}^5}(-6) \longrightarrow \mathcal{O}_{\mathbb{P}^5}(-4)^{\oplus 3} \longrightarrow \mathcal{O}_{\mathbb{P}^5}(-2)^{\oplus 3} \longrightarrow \mathcal{I} \longrightarrow 0 \ ,$$

which readily shows $\mathrm{H}^2(\mathbb{P}^5, \mathcal{I}) = 0$. Hence, X is a K3 surface. □

2. Prove the following theorem of Chern and Griffiths. Let X be a nonsingular surface of degree d in $\mathbb{P}^{n+1}_{\mathbb{C}}$, which is not contained in any hyperplane. If $d < 2n$, then $p_g(X) = 0$. If $d = 2n$, then either $p_g(X) = 0$, or $p_g(X) = 1$ and X is a K3 surface.

Lemma 0.1. *Let H be an ample divisor on a surface X. If D is a divisor on X such that $D.H = 0$ and $\mathrm{H}^0(X, \mathcal{O}_X(D)) \neq 0$, then $\mathcal{O}_X(D) \cong \mathcal{O}_X$.*

Proof. A non-zero section $s \in \mathrm{H}^0(X, \mathcal{O}_X(D))$ defines an effective divisor $D_0 = (s)_0$ linearly equivalent to D. If $D.H = 0$, then $D_0.H = 0$, which is possible if and only if $D_0 = 0$ by (1.10). Hence, $\mathcal{O}_X(D) \cong \mathcal{O}_X$. □

Proof of (Ex. 6.2). Let $\mathcal{O}_X(1)$ be the very ample line bundle on X corresponding to the embedding $X \to \mathbb{P}^{n+1}_{\mathbb{C}}$, let K be a canonical divisor on X, and let $\omega_X = \mathcal{O}_X(K)$ be the canonical sheaf of X. By Bertini's theorem (II, 8.18), a generic hyperplane section of X cuts a smooth irreducible curve $C \subset X$ of degree d with $\mathcal{O}_X(1) \cong \mathcal{O}_X(C)$.

If $d < 2n$, the contrapositive of Clifford's theorem (IV, 5.6) implies $\mathcal{O}_C(1) = \mathcal{O}_X(C) \otimes \mathcal{O}_C$ is a nonspecial line bundle on C, i.e., $h^1(C, \mathcal{O}_C(1)) = 0$. Let ω_C be the canonical sheaf of C. By the adjunction formula (II, 8.20), we have an isomorphism of line bundles

$$\omega_C \cong \omega_X \otimes \mathcal{O}_X(C) \otimes \mathcal{O}_C,$$

so by Serre duality (III, 7.7),

$$h^0(C, \omega_X \otimes \mathcal{O}_C) = h^1(C, \mathcal{O}_X(C) \otimes \mathcal{O}_C) = 0.$$

If there exists a nonzero section $s \in \mathrm{H}^0(X, \omega_X)$, then the zero locus $D = (s)_0$ is an effective divisor of X with finitely many irreducible components. By above, s restricts to zero on any smooth irreducible hyperplane section C, so D must contain C in its support. This is impossible, because there are infinitely many distinct irreducible hyperplane sections of X in $\mathbb{P}^{n+1}_{\mathbb{C}}$. Therefore,

$$p_g(X) = h^0(X, \omega_X) = 0.$$

If $d = 2n$ and $\mathcal{O}_C(1)$ is a special line bundle on C, then by Clifford's theorem, C is a nonhyperelliptic canonical curve of degree $d = C^2 = 2n$ (Ex. 1.2) and genus $g_C = n + 1$, i.e., $\omega_C \cong \mathcal{O}_C(1)$. Combining with the adjunction formula gives the isomorphism

$$\omega_X \otimes \mathcal{O}_X(C) \otimes \mathcal{O}_C \cong \omega_C \cong \mathcal{O}_X(C) \otimes \mathcal{O}_C,$$

which implies

$$\omega_X \otimes \mathcal{O}_C \cong \mathcal{O}_C, \quad C.K = 0.$$

Consider the short exact sequence

$$0 \longrightarrow \mathcal{O}_X(-C) \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{O}_C \longrightarrow 0 \ .$$

Tensoring by ω_X and taking the long exact sequence in cohomology gives

$$0 \longrightarrow \mathrm{H}^0(X, \mathcal{O}_X(K - C)) \longrightarrow \mathrm{H}^0(X, \omega_X) \longrightarrow \mathrm{H}^0(C, \mathcal{O}_C) \ .$$

Since C is very ample and $(K - C).C = -C^2 < 0$, by (1.10) $h^0(X, \mathcal{O}_X(K - C)) = 0$, so by exactness,

$$p_g(X) = h^0(X, \omega_X) \leq h^0(C, \mathcal{O}_C) = 1.$$

If $p_g(X) = 1$, by (0.1) $K \sim 0$. Consider the long exact sequence in cohomology of the short exact sequence above

$$\mathrm{H}^1(X, \mathcal{O}_X(-C)) \longrightarrow \mathrm{H}^1(X, \mathcal{O}_X) \longrightarrow \mathrm{H}^1(C, \mathcal{O}_C) \longrightarrow \mathrm{H}^2(X, \mathcal{O}_X(-C)) \longrightarrow \mathrm{H}^2(\mathcal{O}_X) \longrightarrow 0 \ .$$

By the Kodaira vanishing theorem, $h^1(X, \mathcal{O}_X(-C)) = 0$, and by Serre duality,

$$h^2(X, \mathcal{O}_X(-C)) = h^0(X, \mathcal{O}_X(C)) = n + 2, \quad h^2(X, \mathcal{O}_X) = h^0(X, \mathcal{O}_X) = 1.$$

Since $h^1(C, \mathcal{O}_C) = g_C = n + 1$, we conclude by exactness that $h^1(X, \mathcal{O}_X) = 0$. Hence, X is a K3 surface. □